

PERSONAL INFORMATION

Akin Eker

 Bristol, UK

 akin.eker@bristol.ac.uk | akinneker@gmail.com

 <https://akineker.github.io/>

 www.linkedin.com/in/akin-eker

EDUCATION

2021–2026 **PhD in Computer Science (Awarded 2026)**

Thesis Title: "Blockchain-of-Things: Novel System Architectures and Consensus Models as a Scalable Architecture for IoT-Driven Smart Environments"

Supervisors: Prof. Theo Tryfonas and Prof. George Oikonomou
University of Bristol, UK

2018–2019 **Master of Science in Computer Science**

Thesis Title: "Recognising and Detecting Activities in ADL Datasets using AI"

Supervisor: Dr. Leonardo Aniello
University of Southampton, UK

Taken Modules Include:

– Foundations in Machine Learning | Advanced Machine Learning | Deep Learning | Foundations of Data Science | Data Mining | Data Visualisation

2010–2015 **Bachelor Degree in Computer Engineering**

Dissertation: "Read a Book: A Web Platform that Allows Reading, Purchasing Books, and Note-taking and Researching on One Platform"

Supervisor: Prof. M. Ali Akcayol
Gazi University, Turkey

2012–2013 **Erasmus Programme in Ingeniería Informática (Computer Engineering)**

University of Vigo, Spain

Awarded to students with high GPA and top scores in the English proficiency exam.

PUBLICATIONS AND ATTENDED
CONFERENCES

Conference Papers

- G. Kisa Isik, **A. Eker**, T. Tryfonas, G. Oikonomou, "Security Risk Analysis of Logistical Support Solutions for MaaS and DLT-based Mitigations." *2025 IEEE International Conference on Cyber Security and Resilience (CSR 2025)*, IEEE 2025.
- **Eker, A.**, et al., "SmartMoveChain: A Proposal for Unified Scalable BMaaS for Smart Cities." *2024 IEEE International Smart Cities Conference (ISC2)*. IEEE, 2024.
- Y. Zhang, **A. Eker**, T. Tryfonas, and N. J. Carhart, "Blockchain-based Built Environment Data Management for Modular Construction." *2024 IEEE International Smart Cities Conference (ISC2)*, IEEE, 2024.
- **A. Eker**, T. Tryfonas and G. Oikonomou, "FlexBoT: A Scalable Architecture for Multi-Application Supporting BoT Environments with Application Shifting at Runtime," *2024 IFIP Networking Conference (IFIP Networking)*, Thessaloniki, Greece, 2024, pp. 186-194, doi: 10.23919/IFIPNetworking62109.2024.10619067.
- **A. Eker**, T. Tryfonas and G. Oikonomou, "Public Blockchain in Support of Transactive Energy Markets," *2023 IEEE 9th World Forum on Internet of Things (WF-IoT)*, Aveiro, Portugal, 2023, pp. 01-06, doi: 10.1109/WF-IoT58464.2023.10539475.
- **A. Eker**, T. Tryfonas and G. Oikonomou, "A Conceptual Architecture for Scalable Multi-Application Support in Blockchain-based IoT Environments," *2023 IEEE 21st International Conference on Industrial Informatics (INDIN)*, Lemgo, Germany, 2023.

Journal Papers

- **A. Eker**, T. Tryfonas and G. Oikonomou, "Deconstructing Consensus: A Taxonomy of Protocols, Blockchain Trilemma, System Properties, and Democratic Design", *IEEE Transactions on Network and Service Management* [Under review, Second Round].
- **A. Eker**, T. Tryfonas and G. Oikonomou, "Blockchain-of-Things: A Layered Architecture and Performance Evaluation for Scalable IoT Environments", *Elsevier Internet of Things Journal* [Ready for Submission].

EXPERIENCE

2026 – Present **Research Associate - ELABORATOR Project**

University of Bristol

Collaborating within an international research team on the ELABORATOR project to investigate blockchain and privacy-preserving frameworks for federated learning, and multimodal synthetic data generation approaches.

2022 – 2025 **Teaching Assistant**

University of Bristol, School of Computer Science, and School of Engineering Mathematics and Technology

Assisted Modules Include (Labs):

- **COMSM0049 - Systems and Software Security**
- SEMTM0027 - Statistical Computing and Empirical Methods
- EMATM0067 - Introduction to AI and Text Analytics
- COMSM0089 - Introduction to Data Analytics
- COMS30083 - Machine Learning
- SEMTM0014 - Large Scale Data Engineering with AWS
- SEMTM0028 - Software Development Programming and Algorithms
- EMATM0066 - Visual Analytics
- COMSM0085 - Software Tools
- SEMTM0029 - Financial Technology Group Project with Ethereum

Assisted labs, marked assessments, and supervised individual and group projects.

2017 – 2021 Research and Teaching Assistant*Gazi University, Turkey, Department of Computer Engineering***Given/Assisted Modules Include (Labs and Teaching):**

- **Computer Networks with C++**
- **Embedded Systems with C++**
- Introduction to Artificial Intelligence with Python
- Software Engineering
- Database Management Systems with Oracle
- Data Structures with C++
- Object Oriented Programming with Java
- Model Based Programming in Java

Delivered labs and lectures, supervised individual and group projects, developed course materials, assessed coursework, mentored local and international students. Contributed departmental activities, including representation at academic and industry events, planning departmental tasks, and coordinating their distribution and assignment.

AWARDS & CERTIFICATES**2022 Bristol Plus Award***University of Bristol, UK*

Awarded to only 3% of University of Bristol students each year.

2022 Innovation and Enterprise Certificate*University of Bristol, UK***2016 Turkish Ministry of Education Scholarship for Postgraduate Students***Ankara, Turkey*

Given to students with high GPA and top performance in exams and interviews.

2015 Huawei Seeds for the Future Programme Trainee, Huawei Inc.*Beijing and Shenzhen, China*

Selected as one of 15 students nationwide for training in Huawei's technologies and future vision in Beijing and Shenzhen.

TECHNICAL SKILLS & EXPERTISE**Research and Theoretical Skills:**

- Protocol and Mechanism Design | Consensus Algorithms | Blockchain Architectures | Game Theory | Agent-based Simulation | Artificial Intelligence | Data Mining

Programming Languages:

- Solidity | Rust | Python | C++ | JavaScript

Blockchain Expertise:

- Ethereum (EVM) | Layer 2 Scaling | Custom Rollup Architectures | Zero Knowledge Proofs | zkVM (SP1)

Blockchain Tooling:

- Foundry | Hardhat | Truffle | OpenZeppelin | Chainlink | Infura | Web3.js | Ether.js | Node.js

Data and Infrastructure:

- SQL | No-SQL | IPFS (Pinata) | Docker | Linux

ACADEMIC ENGAGEMENT &
LEADERSHIP

2023 - Present

Peer Reviewer

- 2026 IEEE International Conference on Cyber Security and Resilience (CSR)
- 2025 IEEE International Conference on Cyber Security and Resilience (CSR)
- 2024 Ad-Hoc Networks, Elsevier
- 2023 IEEE International Conference on Communications (ICC)

2026 **Reading Panellist***SparckAI Programme, University of Bristol, UK*

Evaluated prospective student applications for the MSc Artificial Intelligence programme, assessing academic credentials and candidate suitability to ensure high cohort standards.

2022-2025

Chief Resident*University of Bristol, UK*

Facilitated student transition and wellbeing through mentoring, event coordination (open days, end of the year celebrations, etc.), team leadership, and frontline support within residences.

2021-2022

Postgraduate Hall Representative*University of Bristol, UK*

Elected by peers to represent residents, promote wellbeing, and liaise with the university departments.